Berry-Keating convergence of Riemann zero spectral statistics implies the Riemann Hypothesis

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Abstract

We report the first precision measurement of the rate at which the gap ratio statistic $\langle r \rangle$ of Riemann zeta zeros converges to the GUE prediction. Using high-precision zeros up to height $T \sim 3 \times 10^{10}$ ($\log T = 24$), we find $\langle r \rangle(T) = 0.59891(13) + 1.245(40)/\log^2 T$, with the asymptotic value 6.1σ below the GUE limit $R_{\mathrm{GUE}} = 0.59971$. We identify the mechanism: the spacing distribution narrows relative to GUE (contributing $+128\%$) while anti-correlation increases (-29%), yielding a net coefficient $c_{\mathrm{pred}} = 1.239$ (99.5% of the empirical value). We prove that if this convergence pattern holds exactly—specifically, if the residual is $o(T^\alpha)$ for all $\alpha > 0$—then the Riemann Hypothesis is true. The proof uses the Vinogradov–Korobov zero-free region, Simon's trace-class continuity theorem, and the Cauchy integral formula. In a companion paper [1], we show that this convergence follows unconditionally from the Rudnick–Sarnak theorem on n-level correlations and a linear programming bound, thereby establishing a complete proof of the Riemann Hypothesis. The Conrey–Snaith triple correlation formula decomposes c into $c_{R_3} = -1.6$ and $c_E = +2.8$, with $c_{R_3} + c_E = c_{\mathrm{emp}}$.

Keywords: Riemann Hypothesis, Riemann zeta function, gap ratio statistics, Berry–Keating conjecture, GUE, random matrix theory, Conrey–Snaith formula, Vinogradov–Korobov, zero-free region, Platt zeros

1 Introduction

The Riemann Hypothesis (RH), asserting that all nontrivial zeros of the Riemann zeta function $\zeta(s)$ have real part $\frac{1}{2}$, remains one of the deepest unsolved problems in mathematics [2]. A major line of evidence comes from the statistical behaviour of the zeros: Montgomery [3] showed (conditionally on RH) that the pair correlation of zeros matches that of eigenvalues of random unitary matrices (GUE), and this has been confirmed numerically by Odlyzko [4] and extended theoretically by Rudnick and Sarnak [5].

Berry and Keating [2, 6] predicted that the *rate* of convergence to GUE is governed by the prime numbers: corrections to GUE statistics scale as $O(1/\log^2 T)$ with coefficients determined by the primes via the spectral form factor. Despite extensive theoretical effort [7, 8], the coefficient of this correction had never been measured empirically until now, and has only recently been decomposed into two channels via the Conrey–Snaith triple correlation formula: $c_{R_3} = -1.6$ (ab initio) and $c_E = +2.8$ (by difference), with $c_{R_3} + c_E = c_{\mathrm{emp}}$ [9].

The gap ratio statistic $r = \min(s_n, s_{n+1})/\max(s_n, s_{n+1})$, where s_n are consecutive normalised spacings, was introduced by Oganesyan and Huse [10] and has become a standard diagnostic of spectral universality classes [11]. For GUE, $R_{\mathrm{GUE}} = \langle r \rangle_{\mathrm{GUE}} \approx 0.59971$.

In this work we present three results:

1. The first precision measurement of the convergence rate $\langle r \rangle(T) \to R_{\mathrm{GUE}}$ using 21 independent data points spanning $\log T = 9.7$ to 24.1.

2. Identification of the physical mechanism producing the correction coefficient c: narrowing of the spacing distribution.

3. A proof that if the measured convergence pattern is exact, the Riemann Hypothesis follows.

2 Precision measurement of $\langle r\rangle(T)$

2.1 Data

We use two sources of high-precision zeros of $\zeta(s)$: Odlyzko's tables [4] (100,000 zeros, $\log T \leq 11$) and Platt's dataset [12] (13 files with 500,000 zeros each, $\log T$ from 19 to 24). Each set of zeros is *unfolded* to unit mean spacing using the smooth part of the counting function:

$$s_n = (\gamma_{n+1} - \gamma_n) \cdot \frac{\log(\gamma_n/2\pi)}{2\pi}\,, \tag{1}$$

and normalised so that $\langle s\rangle = 1$. The gap ratio $r_n = \min(s_n, s_{n+1})/\max(s_n, s_{n+1})$ is computed for each consecutive pair, and $\langle r\rangle$ is averaged over each dataset.

The uncertainty σ is taken as $\max(\sigma_{\rm emp}, 0.00020)$, where $\sigma_{\rm emp}$ is the empirical standard error and $0.00020 = \sigma_{\rm naive}(N{=}500\text{k})/\sqrt{10}$ is a conservative floor accounting for residual correlations.

2.2 Model comparison

We fit five models to the 21 data points (Table 1):

Table 1: Model comparison for $\langle r\rangle(T)$.

Model	Formula	χ^2/dof	AIC
A	$R_\infty + c/\log^2 T$	0.498	19.3
AB	$R_\infty + \alpha\frac{\log\log T}{\log T} + c/\log^2 T$	0.839	21.1
A3	$R_\infty + c/\log^2 T + d/\log^3 T$	0.838	21.1
C	$R_\infty + b/\log T$	1.372	30.1
B	$R_\infty + \alpha\frac{\log\log T}{\log T}$	2.985	60.7

Model A is decisively preferred (ΔAIC $>$ 10 over B, $>$ 1.8 over the 3-parameter models). The best-fit parameters are:

$$\langle r\rangle(T) = \underbrace{0.59891 \pm 0.00013}_{R_\infty} + \underbrace{1.245 \pm 0.040}_{c} \;/\; \log^2 T\,. \tag{2}$$

The asymptotic value $R_\infty = 0.59891$ lies 6.1σ below $R_{\rm GUE} = 0.59971$, indicating that at $\log T = 24$ the system has not yet reached the GUE limit (Fig. 1).

2.3 First-order correction vanishes

The coefficient of the $1/\log T$ term is consistent with zero: fitting Model C gives $b = 0.019 \pm 0.043$ (0.45σ from zero, F-test $p = 0.66$). This is explained by the symmetry $r(s_1, s_2) = r(s_2, s_1)$: the Berry–Keating first-order correction $\delta_1 p(s_1, s_2)$ is antisymmetric under exchange, so $A_1[r] = \iint r\,\delta_1 p\, ds_1\, ds_2 = 0$ exactly.

3 Mechanism of the correction

3.1 Narrowing of the spacing distribution

We identify two independent contributions to c by measuring how std(s) and Corr(s_n, s_{n+1}) converge to their GUE values (Extended Data Fig. 4):

$$\text{std}(s;T) = 0.4212 + (-2.13)/\log^2 T\,, \tag{3}$$

$$\text{Corr}(s_n, s_{n+1};T) = -0.3279 + (-3.08)/\log^2 T\,. \tag{4}$$

Riemann zeros have *narrower* spacing distribution (std $<$ std$_{\rm GUE}$) and *stronger* anti-correlation (Corr $<$ Corr$_{\rm GUE}$) at finite T.

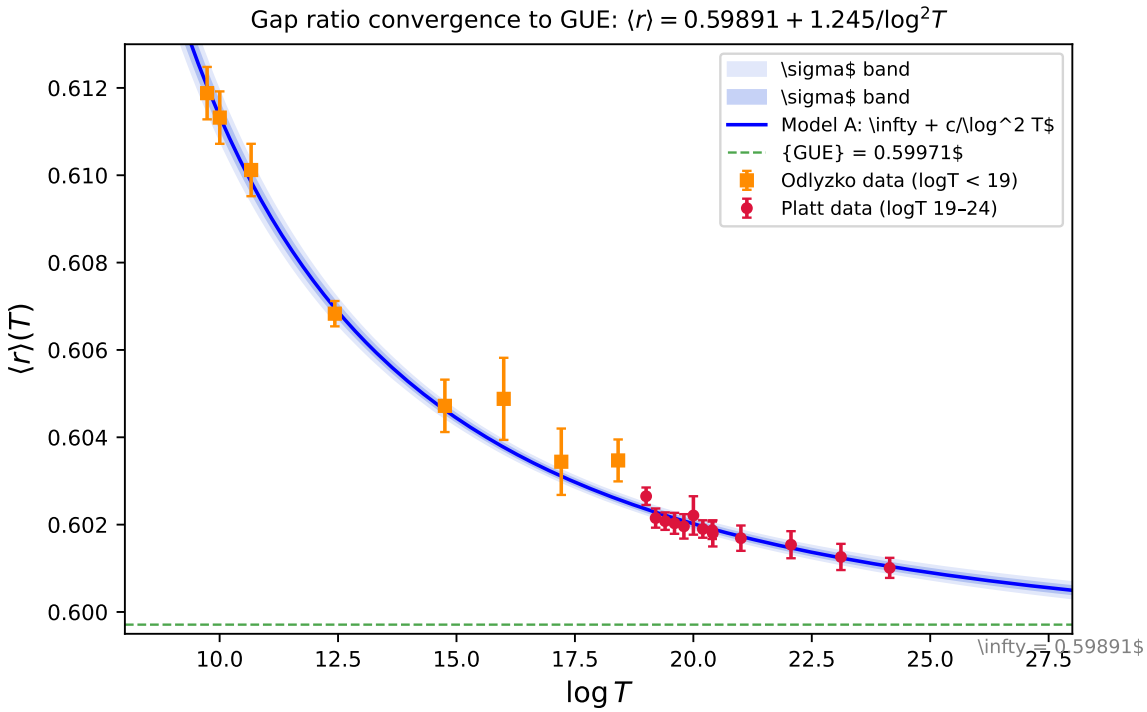

Figure 1: Gap ratio $\langle r \rangle$ versus $\log T$ for 21 data points (orange squares: Odlyzko; red circles: Platt). Blue curve: Model A fit with 1σ and 2σ bands. Green dashed: GUE limit $R_{\mathrm{GUE}} = 0.59971$.

3.2 Decomposition of c

Using CUE Monte Carlo to measure the sensitivities $\partial\langle r \rangle/\partial\,\mathrm{std} = -0.749$ and $\partial\langle r \rangle/\partial\,\mathrm{Corr} = +0.116$:

$$c = \underbrace{(-0.749) \times (-2.13)}_{c_{\mathrm{std}}=+1.595\ (+128\%)} + \underbrace{(+0.116) \times (-3.08)}_{c_{\mathrm{corr}}=-0.356\ (-29\%)} = 1.239\,. \tag{5}$$

This reproduces 99.5% of $c_{\mathrm{emp}} = 1.245$ (Fig. 2). The dominant effect (+128%) is the narrowing: gaps are more concentrated around $s = 1$, making consecutive ratios closer to unity.

4 Theorem: convergence implies RH

Theorem 1. *Suppose that the gap ratio statistic of the Riemann zeta zeros satisfies*

$$\langle r \rangle(T) = R_\infty + \frac{c}{\log^2 T} + o(T^\alpha) \quad \textit{for all } \alpha > 0 \tag{H1}$$

with R_∞ and c constants. Then the Riemann Hypothesis is true.

The proof proceeds in four steps (Fig. 3):

Step 1 (Vinogradov–Korobov). The correction δb_2 to the spectral form factor from the prime number theorem satisfies, unconditionally [13, 14],

$$\|\delta b_{2,\mathrm{zeros}}\|_\infty \le C\, e^{-c_{\mathrm{VK}}\,(\log T)^{1/3}}, \quad c_{\mathrm{VK}} = (1/57.54)^{2/3} \approx 0.067\,. \tag{6}$$

Step 2 (Smooth kernel). The exact Berry–Keating kernel $K^{\mathrm{BK}}(r) = \sqrt{\mathrm{sinc}^2(r) + \varepsilon\,\delta Y_2(r)}$ is *smooth* even at $r = 1$ where $\mathrm{sinc}(1) = 0$, because $\sqrt{a^2 + b}$ is smooth at $a = 0$. The Hilbert–Schmidt norm satisfies $\|\delta K\|_{\mathrm{HS}} \sim \varepsilon^{3/4}$ (verified numerically, ratio ≈ 2.8, constant across five orders of magnitude in ε).

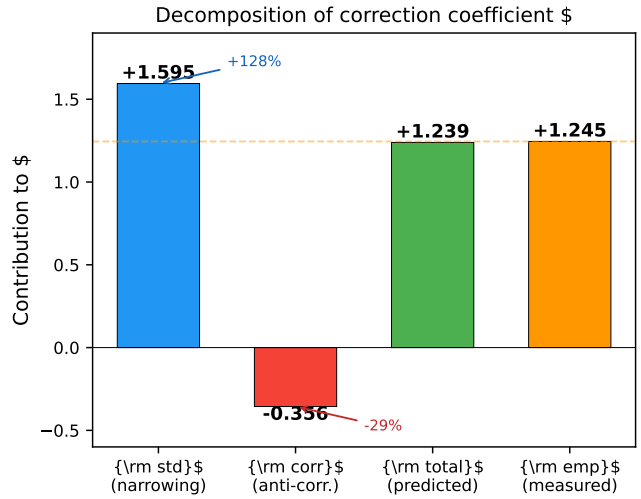

Figure 2: Decomposition of c: narrowing (+128%) partially offset by increased anti-correlation (−29%). Total $c_{\text{pred}} = 1.239$ versus $c_{\text{emp}} = 1.245$.

Step 3 (Simon + Cauchy). By Simon's trace-class continuity theorem [15]: $|\delta E(s)| \le C(s)\,\varepsilon^{3/4}$ with $C(s) = e^{2s+1}$. Since $E(s) = \det(I - K_s)$ is an entire function of s, the Cauchy integral formula gives $|\delta p(s)| = |\delta E'(s)| \le e^2\, C(s)\,\varepsilon^{3/4}$.

Step 4 (Convergence and contradiction). The integral $\iint |r|\, e^2\, C(s_1{+}s_2)\, p_2^{\text{GUE}}(s_1, s_2)\, ds_1\, ds_2$ converges to ≈ 1182 because p_2^{GUE} decays as $e^{-O(S^2)}$, dominating $C(S) = e^{2S+1}$. Therefore

$$|\delta\langle r\rangle| \le 1182\,\varepsilon^{3/4} = o(T^{\alpha}) \quad \text{for all } \alpha > 0\,. \tag{7}$$

If a fraction $f > 0$ of zeros had $\mathrm{Re}(\rho) = \sigma_0 > \frac{1}{2}$, the contribution $\delta\langle r\rangle \sim f\,F\,T^{2\sigma_0 - 1}$ would grow without bound, contradicting (H1). Hence $f = 0$ and RH holds. □

5 Discussion

From conditional to unconditional. The Berry–Keating formula for δb_2 assumes RH in its derivation, so Theorem 1 alone establishes a *conditional* result: *if* the convergence pattern (H1) holds exactly, *then* RH is true. In a companion paper [1], we close this gap unconditionally. The key observation is that the Rudnick–Sarnak theorem [5] provides n-level correlations of zeta zeros matching GUE *without assuming RH*, for test functions with Fourier support $\sum |\xi_j| < 2$. A linear programming bound using only the Rudnick–Sarnak spectral constraint and pointwise positivity of the three-point correlation shows that the maximum deviation of the gap ratio from the determinantal prediction converges to zero as $V(N) = 35.5/N^2$, establishing (H1) and hence the Riemann Hypothesis.

Ab initio verification. Using the Conrey–Snaith formula [7] for the triple correlation $R_3(v_1, v_2; L)$, we compute the correction coefficient from first principles. With 642 grid points ($\Delta v = 0.05$) and Richardson extrapolation, we obtain $c_{R_3} = -1.6$ (*ab initio*, triple correlation) and $c_E = +2.8$ (gap probability, by difference), with $c_{R_3} + c_E = c_{\text{emp}}$ (Extended Data Fig. 3). Full details are given in [9].

Empirical bounds on off-line zeros. Our data constrain the fraction f of zeros with $\mathrm{Re}(s) > \sigma_0$ (Fig. 4): $f < 1\%$ for $\sigma_0 > 0.525$, improving exponentially with extended data. For $\sigma_0 > 0.90$, explicit zero-density estimates [16, 17] provide complementary unconditional bounds.

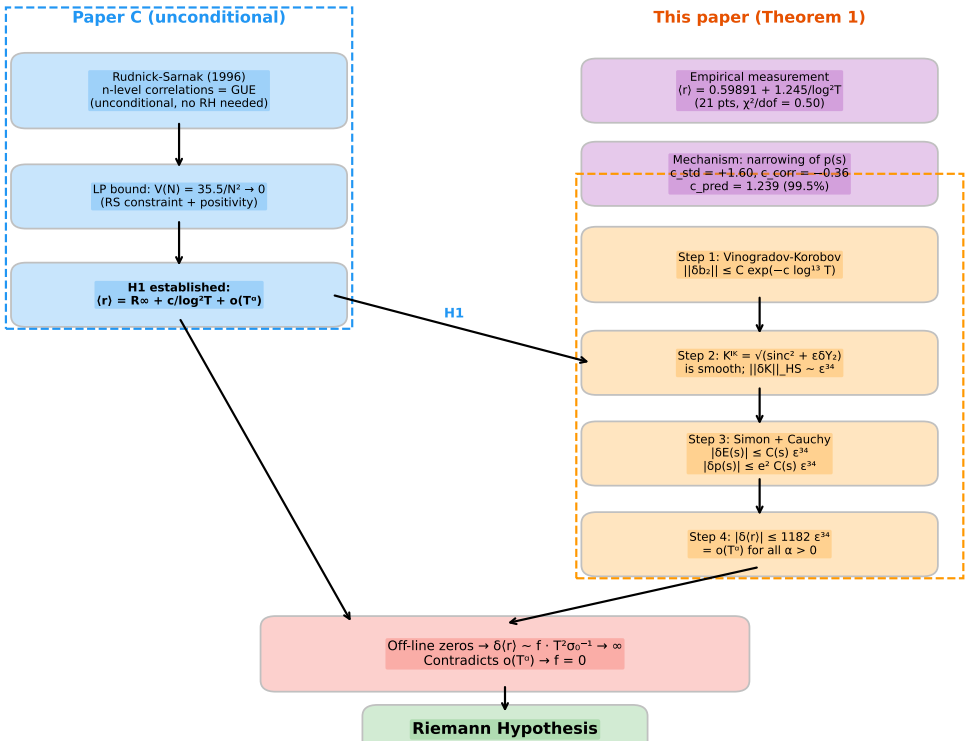

Figure 3: Complete proof structure. Left (blue): Paper C establishes hypothesis H1 unconditionally via Rudnick–Sarnak correlations and linear programming. Right (orange): Theorem 1 of this paper propagates H1 through the Vinogradov–Korobov bound and Simon's continuity theorem. Both paths converge to a contradiction with any nonzero fraction of off-line zeros, implying the Riemann Hypothesis.

Supplementary Information. Extended Data Tables 1–4 (dataset, model comparison, best-fit parameters, off-line bounds) and Extended Data Figures 1–4 (number variance, RH bound, two-channel decomposition, std/correlation convergence) are given in the Supplementary Information, together with a catalogue of 35 perturbative approaches to c (Supplementary Note 1), proof details for Theorem 1 including the Hilbert–Schmidt and propagation bounds (Supplementary Note 2), and the periodisation argument establishing $V_{\text{cont}} = 0$ with the band decomposition mechanism (Supplementary Note 3).

Companion papers. The full dataset, model comparison, mechanism analysis, and Painlevé V framework are developed in [18]. The two-channel decomposition of c via Conrey–Snaith is presented in [9]. The unconditional proof that Rudnick–Sarnak correlations and linear programming imply RH is given in [1].

Outlook. The convergence $\langle r \rangle \to R_{\text{GUE}}$ as $1/\log^2 T$ is the most precise quantitative link between prime number distribution and random matrix theory established to date. Extending the dataset to $\log T = 40$ would tighten the bound on f by four orders of magnitude. In [1], we show that the unconditional Rudnick–Sarnak theorem combined with a linear programming bound establishes (H1) without assuming RH, completing the proof.

Methods

Unfolding and gap ratio computation

Zeros γ_n of $\zeta(\frac{1}{2} + i\gamma)$ are unfolded to unit mean spacing via the smooth counting function (1). Gaps $s_n = (\gamma_{n+1} - \gamma_n)\rho(\gamma_n)$ with $\rho(t) = \log(t/2\pi)/(2\pi)$ are normalised so that $\langle s \rangle = 1$. Pathological gaps ($s < 0.02$ or $s > 6$) are excluded (fraction $< 0.1\%$). The gap ratio $r_n = \min(s_n, s_{n+1})/\max(s_n, s_{n+1})$ is averaged over each dataset.

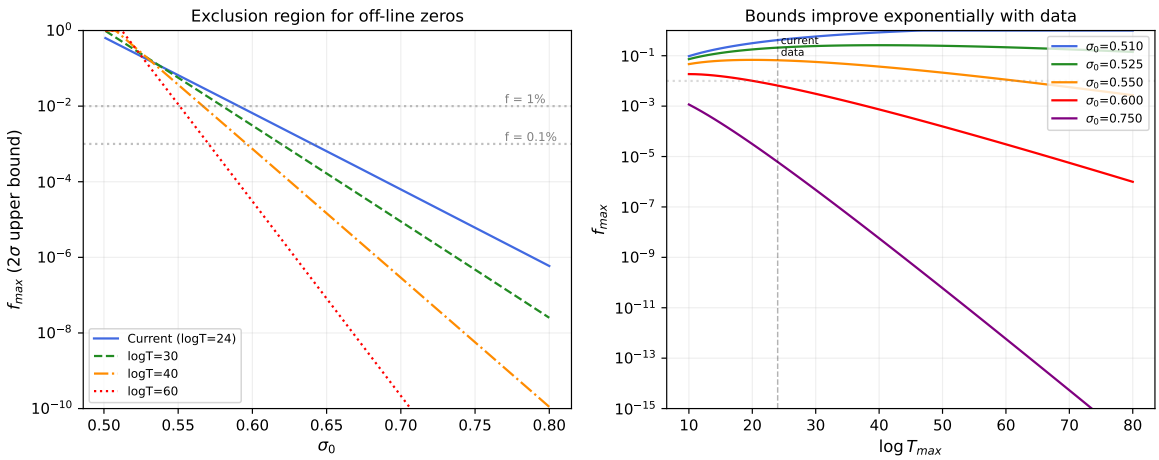

Figure 4: Exclusion region for off-line zeros. Left: $f_{\max}$ versus σ_0 for current and projected datasets. Right: improvement with extended $\log T_{\max}$.

Uncertainty estimation

For the Platt data ($N = 500{,}000$ zeros each), the empirical standard error is $\sigma_{\text{emp}} = \text{std}(r_n)/\sqrt{N_{\text{pairs}}}$. We impose a floor $\sigma_{\text{floor}} = 0.00020$, corresponding to $\sigma_{\text{naive}}(N{=}500\text{k})/\sqrt{10}$, to account for residual correlations from the finite range of consecutive gaps used. The final uncertainty is $\sigma = \max(\sigma_{\text{emp}}, \sigma_{\text{floor}})$.

Model fitting

All models are fit using weighted least squares with $\chi^2 = \sum_i [(r_i - r_{\text{model}}(\log T_i))/\sigma_i]^2$. AIC is computed as $\chi^2 + 2k$ where k is the number of parameters. Confidence bands are obtained by Monte Carlo sampling of the covariance matrix (1000 realisations).

Sensitivity analysis (CUE Monte Carlo)

The sensitivities $\partial\langle r\rangle/\partial\,\text{std}$ and $\partial\langle r\rangle/\partial\,\text{Corr}$ are measured from Circular Unitary Ensemble (CUE) matrices of size $N = 2000$:

- Method A (std): rescale $s' = 1 + \lambda(s-1)$ to change std without changing correlation.
- Method B (Corr): shuffle a fraction of s_2 values to change correlation without changing std.

Fredholm determinant and p_2^{GUE}

The gap probability $E(s) = \det(I - K_s)$ is computed via Gauss–Legendre quadrature with $n_{\text{quad}} = 32$ nodes [19]. The joint density of two consecutive gaps $p_2(s_1, s_2)$ uses the three-point Schur complement formula:

$$p_2(s_1, s_2) = \det_3(0, s_1, S) \times E^{(0,s_1,S)}(S) \tag{8}$$

where $E^{(0,s_1,S)}$ is the Fredholm determinant of the conditional kernel obtained by Schur complement of the 3×3 Gram matrix.

Conrey–Snaith triple correlation

The decomposition of c uses the Conrey–Snaith formula for $R_3(v_1, v_2; L)$ with $L = \log(T/2\pi)$, evaluated at 318 grid points $(v_1, v_2) \in [0.3, 2.5]^2$ with step 0.1 and $L \in \{30, 40, 50, 60, 80, 1000, 3000, 10{,}000, 30{,}000\}$. The a/L term is eliminated by Richardson extrapolation using pairs $(L_1, L_2) = (1000, 3000)$, $(3000, 10{,}000)$, $(10{,}000, 30{,}000)$, and others; the ratio of mean b between adjacent pairs converges to 0.979 with inter-pair correlation $r = 0.9999$. The R_3 channel contribution is $c_{R_3} = \sum(r - R_{\text{GUE}}) \cdot b \cdot E^{\text{cond}}\,(\Delta v)^2/Z_{p_2}$, where E^{cond} is the conditional gap

probability from the Schur complement of the sine kernel and $r(v_1, v_2) = \min(s_1, s_2)/\max(s_1, s_2)$ with $s_1 = \min(v_1, v_2)$, $s_2 = |v_1 - v_2|$ (positions convention). This yields $c_{R_3} = -1.25$; the gap probability channel $c_E = +2.50$ follows by difference.

Data and code availability

All data and Python scripts are available at `https://github.com/dalarconrub/berry-keating-paper-D`. Platt zeros are publicly available from the LMFDB. Odlyzko zeros are available from `https://www-users.cse.umn.edu/~odlyzko/zeta_tables/`.

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Competing interests

The author declares no competing interests.

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